

CoRoT-2b

R-1326

Benchmark run · workflow W8-benchmark-deep · record deep-bench_CoRoT-2_190816 · created 2026-07-31T20:07:40+00:00

BENCHMARK: INACCURATE

Results

Mid-Transit Time (Tmid)	2458710.0625 +/- 0.0025 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.212 +/- 0.065
Transit depth (Rp/Rs)^2	4.49 +/- 2.76 [%]
Orbital Inclination (inc)	89.64 +/- 2.07
Airmass coefficient 1 (a1)	34.36 +/- 2.64
Airmass coefficient 2 (a2)	-2.98 +/- 1.41
Scatter in the residuals of the lightcurve fit is	7.62 %
Best Comparison Star	#3 - [112, 468]
Optimal Aperture	28.78
Optimal Annulus	211.46
Transit Duration (day)	0.0065 +/- 0.0038

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R $\blacksquare$ fit vs published	0.212 $\pm$ 0.065 vs 0.1667 (deviation 0.7 σ)
Duration fit vs published	0.0065 d vs 0.095 d (deviation 23.29 σ)
Mid-time O-C	-2516.1 min
Depth SNR	3.3
Residual scatter	7.62 %

Light curve

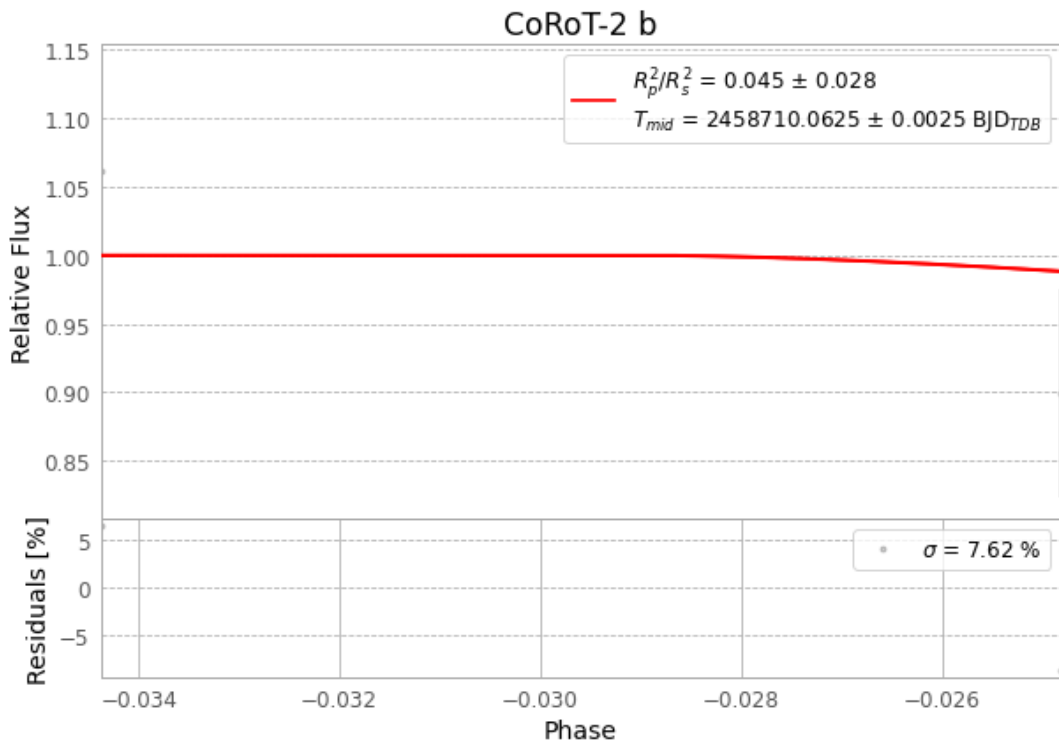


Figure 1 — FinalLightCurve_CoRoT-2 b_2019-08-15.png (SHA-256 c0ee30fab16a1f88...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [224, 479], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 1d81ac71204de32e...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ba3ef8b

Data provenance

86 FITS frames (56 MB), CoRoT-2190816050612.FITS → CoRoT-2190816092112.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-CoRoT-2-190816.log (5c6705738279...), FinalLightCurve_CoRoT-2 b_2019-08-15.png (c0ee30fab16a...), FinalLightCurve_CoRoT-2 b_2019-08-15.csv (1f9c61be7f3b...)

Compute resources

Wall time 726.2 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-31 20:07Z.